

Contents

1	Quadratic Forms and Spaces	2
2	Proof of the Hasse-Minkowski Theorem	4
2.1	$n = 2$	4
2.2	$n = 3$	5
2.3	$n = 4$	6
2.4	$n \geq 5$	8
3	Number-Theoretic Applications	9
4	A Cohomological Interpretation due to C. Lane	12

List of Symbols

(V, Q)	Quadratic Space over K with vector space V and quadratic form Q	2
V_v	$V \otimes_K K_v$	3
I_K	Idèles over K	6
C_L	Idèle class group over L	6
$H^i(L/K)$	$H^i(\text{Gal}(L/K), L^\times)$	6
$O(K, \varphi)$	Orthogonal group of K w.r.t. φ	12

Preface

These notes were made for a seminar class on number-theoretic local-to-global principles. My thanks to [Dr. Tim All](#) for supervising the seminar and to [Connor Lane](#) for making the seminar much more enjoyable and fruitful. I would also like to thank Connor for providing the cohomological point of view given in the last section. My thanks to Keith Bauer for pointing out some errors.

Unless otherwise noted, all fields have characteristic $\neq 2$. For more reading, see [4] and [2].

One of the classical local-to-global results in number theory is the *Hasse-Minkowski Theorem*, which states the following:

Theorem 1: Hasse-Minkowski

Let Q be a quadratic form over a global field K ($\text{char } K \neq 2$). Then Q represents 0 nontrivially over K ($Q(\mathbf{x}) = 0$ for some $\mathbf{x} \neq \mathbf{0}$) if and only if Q is *isotropic* (represents 0 nontrivially) over all completions K_v of K .

Of course, one direction is clear; if we have a nontrivial solution to $Q(\mathbf{x}) = 0$ over K , then we also have nontrivial solutions in every completion by the natural embedding $K \hookrightarrow K_v$. The converse, sadly (or happily, depending on your point of view), takes a bit more effort to prove.

1 Quadratic Forms and Spaces

We begin with a review of the basic algebraic theory of quadratic forms.

Quadratic Forms

A *quadratic form* on a finite dimensional vector space V of K is a function $Q : V \rightarrow K$ such that

1. $Q(c\mathbf{v}) = c^2Q(\mathbf{v})$
2. The pairing $\langle \cdot, \cdot \rangle : V \times V \rightarrow K$ given by

$$\langle \mathbf{v}, \mathbf{w} \rangle := \frac{1}{2} (Q(\mathbf{v} + \mathbf{w}) - Q(\mathbf{v}) - Q(\mathbf{w}))$$

is bilinear.

Any quadratic form is given by a homogeneous polynomial of degree 2 in $\dim V$ variables once a basis for V is fixed. It will also be convenient to express $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where $A \in M_n(K)$; for extra convenience we may take A to be a symmetric matrix. The choice of A as symmetric gives Q a canonical matrix representation.

We say two forms P and Q are equivalent if there exists $M \in GL_n(K)$ such that $P(M\mathbf{x}) = Q(\mathbf{x})$. If A and B are the defining symmetric matrices of P and Q , this is the statement that $A = M^T B M$ for $M \in GL_n(K)$.

We say a quadratic form is *non-degenerate* if $\text{disc}(Q) := \det(A)$ is nonzero. For diagonal forms in particular, this means that $Q(\mathbf{x}) = a_1 x_1^2 + \cdots + a_n x_n^2$ where $a_i \neq 0$ for all i . Note that if P and Q are equivalent forms, then $\det(A) = \det(M^T B M) = \det(B) \det(M)^2$. This means the discriminant of an equivalence class of quadratic forms is defined up to its class modulo $(K^\times)^2$.

Quadratic Space

A *quadratic space* over K is a pair (V, Q) where V is a vector space over K and $Q : V \rightarrow K$ is a quadratic form. We may use the pairing $\langle \cdot, \cdot \rangle$ associated to Q to view (V, Q) as a sort of inner product space over K . In particular, $\langle \mathbf{v}, \mathbf{v} \rangle = Q(\mathbf{v})$ for any $\mathbf{v} \in V$ by definition.

The following proposition is central to many arguments involving quadratic forms. In particular, we will use this to reduce the proof of **theorem 1** to the case of diagonal quadratic forms; that is, we have nontrivial solutions to $Q(\mathbf{x}) = 0$ if and only if we have nontrivial solutions to an equivalent diagonal form.

Proposition 1

Let (V, Q) be a quadratic space over K (where $\text{char } K \neq 2$). Then Q is *equivalent to a diagonal form* i.e.

$$Q(\mathbf{x}) = \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2 \quad \text{where} \quad \mathbf{x} = \sum_{i=1}^n x_i e_i$$

for some basis $\{e_1, \dots, e_n\}$ of V .

Exercise 1

Prove **proposition 1**.

We define the direct sum of quadratic spaces (V, P) and (W, Q) to be the vector space $V \oplus W$ with the form $A(\mathbf{x} \oplus \mathbf{y}) = P(\mathbf{x}) + Q(\mathbf{y})$ for $\mathbf{x} \in V, \mathbf{y} \in W$. We have the following:

Proposition 2

Suppose (V, Q) is a non-degenerate quadratic space over K . The following are equivalent for $\alpha \in K^\times$:

1. $Q(\mathbf{x}) = \alpha$ has a solution over K (α is *represented* by Q)
2. $Q(\mathbf{x}) - \alpha y^2$ is isotropic over K
3. $(V, Q) = (K, \alpha x^2) \oplus (V', Q')$.

Exercise 2

Prove **proposition 2**.

We note an important consequence of this proposition: if Q is isotropic over K , then every $\alpha \in K$ is represented by Q . **Proposition 2** also allows us to state a local-to-global property for representability by quadratic forms:

Theorem 2: Hasse Principle for Representability

Let Q be a quadratic form over a global field K ; $\alpha \in K$ is represented by Q if and only if α is represented by Q over K_ν for all completions K_ν of K .

Proof. This follows immediately from **proposition 2** and the Hasse-Minkowski Theorem (**theorem 1**). ■

We say that (V, P) *embeds isometrically* into (W, Q) if there is an injective linear map $T: V \rightarrow W$ such that $P(\mathbf{x}) = Q(T\mathbf{x})$ for all $\mathbf{x} \in V$. We prove a consequence of the Hasse-Minkowski Theorem for embeddings of quadratic spaces:

Theorem 3: Hasse Principle for Quadratic Space Embeddings

Let (V, P) and (W, Q) be quadratic spaces over a global field K . The following are equivalent:

1. There is an isometric embedding $(V, P) \hookrightarrow (W, Q)$.
2. There is an isometric embedding $(V_\nu, P) \hookrightarrow (W_\nu, Q)$ for all places ν of K , where we write V_ν to mean $V \otimes_K K_\nu$ (and similar for W).

We note an important special case of this theorem where $\dim V = \dim W$:

Theorem 4: Hasse Principle for Quadratic Space Isomorphisms

The following are equivalent:

1. There is an isomorphism (bijective isometric map) $(V, P) \xrightarrow{\sim} (W, Q)$.
2. There is an isomorphism $(V_\nu, P) \xrightarrow{\sim} (W_\nu, Q)$ for all places ν of K .

When $V = W$ (so $T \in \text{Aut}_K(V)$), this tells us that two quadratic forms are equivalent over K if and only if they are equivalent over all completions K_ν .

Theorem 5: Witt Cancellation

Suppose $(U, R), (V, P), (W, Q)$ are non-degenerate quadratic spaces (the forms are non-degenerate) over K and suppose $(U, R) \oplus (V, P) \cong (U, R) \oplus (W, Q)$ (where $\cong$ denotes isometric isomorphism as in **theorem 4**). Then we can ‘cancel’ (U, R) from the isomorphism i.e. $(V, P) \cong (W, Q)$.

Proof. The theorem will follow from the case where U is one-dimensional and the fact that every quadratic form is diagonalizable (**proposition 1**). To that end, suppose without loss of generality that we have $R = a_1 x_1^2$, $P = a_2 x_2^2 + \cdots + a_n x_n^2$, and $Q = b_2 y_2^2 + \cdots + b_n y_n^2$ (isomorphic quadratic spaces are necessarily of the same dimension). Then we have the identity

$$R \oplus P(\mathbf{x}) = a_1 x_1^2 + a_2 x_2^2 + \cdots + a_n x_n^2 = R \oplus Q(M\mathbf{x}) = R \oplus Q(\mathbf{y}) = a_1 y_1^2 + b_2 y_2^2 + \cdots + b_n y_n^2 \quad (1)$$

where $M \in GL_n(K)$. In particular, we have that $y_i = m_{i1}x_1 + \cdots + m_{in}x_n$ ($m_{ij} \in K$) for each $i = 1, \dots, n$. We now consider the substitution $x_1 \mapsto \frac{z}{1-m_{11}}$ where $z = y_1 - m_{11}x_1 = m_{12}x_2 + \cdots + m_{1n}x_n$. If $m_{11} = 1$, we can take m_{1j} to $-m_{1j}$ and thus y_1 to $-y_1$, which changes nothing in the identity equation (1). Thus, we can suppose $m_{11} \neq 1$ so the substitution is well-defined. We now have

$$a_1x_1^2 = a_1 \left(\frac{z}{1-m_{11}} \right)^2 = a_1 \left(z + \frac{m_{11}z}{1-m_{11}} \right)^2 = a_1(z + m_{11}x_1)^2 = a_1y_1^2.$$

This gives us the equation

$$a_1x_1^2 + a_2x_2^2 + \cdots + a_nx_n^2 = a_1y_1^2 + b_2y_2^2 + \cdots + b_ny_n^2$$

from which we can cancel the terms $a_1x_1^2 = a_1y_1^2$ to obtain

$$a_2x_2^2 + \cdots + a_nx_n^2 = b_2y_2^2 + \cdots + b_ny_n^2$$

where each y_i is now expressed as a linear combination of $x_2, \dots, x_n$ (since the substitution expressed x_1 as a linear combination of $x_2, \dots, x_n$). It remains to show that this linear transformation is invertible. To that end, let $A = [a_{ij}]$ be the matrix corresponding to this transformation. Then the transformation of quadratic forms is given by $A = N^TBN$ where $A = \text{diag}\{a_2, \dots, a_n\}$ and $B = \text{diag}\{b_2, \dots, b_n\}$. Since A is non-degenerate, $\det(A) \neq 0$, from which it follows that $\det(N) \neq 0$ as well. ■

Proof of theorem 3. 1. $\Rightarrow$ 2. is immediate. For the reverse implication we induct on dimension of V . The case $\dim V = 0$ is trivial, so we suppose now that the theorem holds for dimension n . We can write $(V, P) = (K, \alpha x^2) \oplus (V', P')$. Then by assumption we have that for each place ν of K ,

$$(K_\nu, \alpha x^2) \hookrightarrow (V_\nu, P) \hookrightarrow (W_\nu, Q)$$

so α is represented by W_ν . Then it follows that the space $(W_\nu, Q) \oplus (K, -\alpha x^2)$ is isotropic. The Hasse-Minkowski Theorem implies that $(W, Q) \oplus (K, -\alpha x^2)$ is isotropic over K , so we can write $(W, Q) = (K, \alpha x^2) \oplus (W', Q')$. Now, we can write

$$(K_\nu, \alpha x^2) \oplus (W'_\nu, Q') = (W_\nu, Q) = (V_\nu, P) \oplus (X_\nu, R) = (K_\nu, \alpha x^2) \oplus (V'_\nu, P') \oplus (X_\nu, R).$$

We apply the Witt Cancellation theorem (**theorem 5**), which gives us that $(V'_\nu, P') \oplus (X_\nu, R) \cong (W'_\nu, Q')$, so (V'_ν, P') embeds isometrically into (W'_ν, Q') for all places ν of K . By induction hypothesis, (V', P') embeds isometrically into (W', Q') , which naturally extends to an isometric embedding $(V, P) \hookrightarrow (W, Q)$. ■

2 Proof of the Hasse-Minkowski Theorem

We will prove the Hasse-Minkowski theorem by first proving it for quadratic forms in two variables ($n = 2$); we will then prove the theorem for $n = 3$, then $n = 4$, and finally for $n \geq 5$. We use **proposition 1** to reduce to the case where the form is diagonal. We also assume forms are non-degenerate, as isotropy in this case is trivial.

We begin with the simplest case:

2.1 $n = 2$

Proposition 3: Binary Hasse-Minkowski

The Hasse-Minkowski theorem holds for quadratic forms in two variables.

Proof. Let $Q(x, y) = ax^2 + by^2$; if we have nontrivial zeros then both x and y must be nonzero. Rearranging, we get that $\left(\frac{x}{y}\right)^2 = -\frac{b}{a}$; this means if we have a nontrivial zero in K_ν , then $-b/a$ is a square in K_ν . On the other hand, if

$-b/a = \alpha^2$ is a square in K , then we can see that $ax^2 + by^2 = 0$ has a nontrivial solution if and only if

$$x^2 + \frac{b}{a}y^2 = x^2 - (\alpha y)^2 = (x + \alpha y)(x - \alpha y)$$

has a nontrivial zero; taking $(x, y) = (\alpha, 1)$ produces such a zero. We have thus reduced the binary case of Hasse-Minkowski to a local-to-global question about squares: if $\alpha \in K^\times$ is a square in all completions $K_\nu^\times$, then is it a square in $K^\times$ itself? The answer to this question is affirmative per the following theorem:

Theorem 6: Local-to-Global for Squares

Let K be a global field; for $\alpha \in K^\times$ we have $\alpha \in (K^\times)^2$ if and only if $\alpha \in (K_\nu^\times)^2$ for all completions K_ν of K .

Proof. Clearly if α is a square in K , then it is a square in all K_ν . Conversely, suppose that $\alpha \in K^\times$ is not a square. By the Chebotarev Density theorem, the finite places which do not split in $K(\sqrt{\alpha})$ have density $\frac{1}{2}$. In particular, such places exist and for any one of them we have $K_\nu \otimes_K K(\sqrt{\alpha}) = K_\nu(\sqrt{\alpha})$ is a field. Thus α is not a square in these K_ν . ■

2.2 $n = 3$

We now progress to the ternary form of **theorem 1**:

Proposition 4: Ternary Hasse-Minkowski

The Hasse-Minkowski Theorem holds for ternary quadratic forms.

Proof. We examine the form $Q(x, y, z) = ax^2 + by^2 + cz^2$; dividing by c shows that the form is equivalent to $\frac{a}{c}x^2 + \frac{b}{c}y^2 + z^2$, so we can consider the case where $c = 1$. Rearranging the equation $ax^2 + by^2 + z^2 = 0$ gives us

$$-a = \left(\frac{z}{x}\right)^2 + b\left(\frac{y}{x}\right)^2 = \text{Norm}_{K(\sqrt{-b})/K}\left(\frac{z}{x} + \sqrt{-b}\frac{y}{x}\right)$$

so if $Q(x, y, z) = ax^2 + by^2 + cz^2$ has a nontrivial zero in K_ν , then $-a$ is in the norm group $\text{Norm}_{K_\nu(\sqrt{-b})/K_\nu}(K_\nu(\sqrt{-b})^\times)$. Similarly to the proof of **proposition 3**, we can go the other way; suppose $-a = \text{Norm}_{K(\sqrt{-b})/K}(\alpha + \beta\sqrt{-b})$ for some $\alpha, \beta \in K$. Then we have

$$ax^2 + by^2 + z^2 = -(\alpha^2 + b\beta^2)x^2 + by^2 + z^2 = b(y^2 - \beta^2x^2) + (z^2 - \alpha^2x^2).$$

Choosing $(x, y, z) = (1, \beta, \alpha)$ then gives a nontrivial zero for the form.

As in the proof of **proposition 3**, we have reduced the statement of ternary Hasse-Minkowski to a local-to-global statement about norms: if $a \in K^\times$ is a local norm in the extension L/K for all completions K_ν of K , then is a a global norm? The answer to this question is yes for cyclic extensions, by the Hasse Norm Theorem (**theorem 7**). Thus, ternary Hasse-Minkowski will be implied by the next theorem. ■

Theorem 7: Hasse Norm Theorem

Suppose L/K is a cyclic extension of global fields. Then $\alpha \in K^\times$ is a norm from $L^\times$ if and only if it is a local norm (i.e. $\alpha \in \text{Norm}_{L_\nu/K_\nu}(L_\nu^\times)$ where $L_\nu = L \otimes_K K_\nu$) for all completions K_ν of K .

Proof. We want injectivity of the map

$$K^\times / \text{Norm}_{L/K}(L^\times) \longrightarrow \prod_\nu K_\nu^\times / \text{Norm}_{L_\nu/K_\nu}(L_\nu^\times).$$

Note that $L_v = \prod_{w|v} L_w$ so we should account for this by taking K_v with multiplicity $r(v) = [L : K]/e(v)f(v)$ in the product. To treat this, we note that the image of $K^\times$ in $\prod_v K_v^\times$ is contained in I_K , the idèles over K ; thus, it suffices to show injectivity of the map

$$K^\times / \text{Norm}_{L/K}(L^\times) \longrightarrow I_K / \text{Norm}_{L/K}(I_L)$$

induced from cohomology. This is slightly different than what we started with, since $\text{Norm}_{L/K}(\alpha)_v = \prod_{w|v} \text{Norm}_{L_w/K_v}(\alpha_w)$. However, it follows that if $\gamma \in \text{Norm}_{L_v/K_v}(L_v^\times)$, then $\gamma \in \text{Norm}_{L/K}((I_L)_v)$. Indeed, if $\gamma = \text{Norm}_{L_{v_i}/K_v}(\beta_i)$, then take the idèle $(\alpha) \in I_L$ with β_i in the v_i -component and 1 in all other components above v . Then $\text{Norm}_{L/K}(\alpha)_v = \gamma$.

Now recall that we have the short exact sequence

$$1 \longrightarrow L^\times \longrightarrow I_L \longrightarrow C_L \longrightarrow 1$$

which defines C_L , the idèle class group. Taking cohomology, we have the long exact sequence

$$H^1(L/K) \rightarrow H^1(L/K, I_L) \rightarrow H^1(L/K, C_L) \rightarrow H^2(L/K) \rightarrow H^2(L/K, I_L) \rightarrow H^2(L/K, C_L)$$

where we write $H^i(L/K)$ to mean $H^i(L/K, L^\times)$. It is an important result of class field theory that $H^1(L/K, C_L) = 1$ (see any book on class field theory, for example Chapter VI of [1]). This tells us that the map

$$H^2(L/K) \longrightarrow H^2(L/K, I_L)$$

is injective. However, for L/K a cyclic extension, we have that $H^2(L/K) = \hat{H}^0(L/K) = K^\times / \text{Norm}_{L/K}(L^\times)$ and $H^2(L/K, I_L) = I_K / \text{Norm}_{L/K}(I_L)$. As mentioned previously, if $\alpha \in \text{Norm}_{L_v/K_v}(L_v^\times)$ then $\alpha \in \text{Norm}_{L/K}(I_L)$. Then α defines the zero class in $I_K / \text{Norm}_{L/K}(I_L)$ and thus the zero class in $K^\times / \text{Norm}_{L/K}(L^\times)$. This proves the theorem. ■

Remark:

In the language of Tate-Shafarevich groups, we have proven the following for L/K a cyclic extension of global fields:

$$\hat{\text{III}}^0(L/K, L^\times) \cong \text{III}^{2i}(L/K, L^\times) = 1$$

2.3 $n = 4$

It is a curious thing that in the classical proof of Hasse-Minkowski over $\mathbf{Q}$, the $n = 3$ has little technical difficulty, while the $n = 4$ case requires calls to Hilbert reciprocity and Dirichlet's theorem on primes in arithmetic progressions. Of course at this point, we should have no qualms about using either of these tools (or Chebotarev Density Theorem in place of Dirichlet); nevertheless, it turns out we can get away with much less machinery in proving quaternary Hasse-Minkowski over all global fields. Why do we need such machinery as Hilbert reciprocity and Dirichlet's theorem when proving the theorem over $\mathbf{Q}$ then? The proof we give here proceeds by first proving a special case of quaternary Hasse-Minkowski as a consequence of the ternary theorem; we then make an argument whereby we pass to an appropriate quadratic extension of K over which our form is forced to be isotropic. We finally deduce isotropy over K from the isotropy over this extension. The obstruction in the proof over $\mathbf{Q}$ is now clear: in order to replicate this argument, one would need to prove ternary Hasse-Minkowski for *all* quadratic number fields. However, this seems rather intractable without some amount of fancy machinery from algebraic number theory, if not a good treatment of class field theory over $\mathbf{Q}$.

We first prove the aforementioned special case of quaternary Hasse-Minkowski; namely, where the form has discriminant in $(K^\times)^2$. (Recall that the discriminant of a diagonal quadratic form is the product of its coefficients.) We make use of the following lemma:

Lemma 1

The following are equivalent:

- (i) The form $x^2 - ay^2 - bz^2$ is isotropic over k .
- (ii) The form $x^2 - ay^2 - bz^2 + abw^2$ is isotropic over k .
- (iii) $b \in \text{Norm}_{k(\sqrt{a})/k}(k(\sqrt{a})^\times)$.

Proof. (i) implies (ii) trivially; set $w = 0$. As discussed in the proof of [proposition 4](#), (i) and (iii) are equivalent. Thus, it remains to show that (ii) implies (iii).

Suppose $x^2 - ay^2 - bz^2 + abw^2 = 0$ for some x, y, z, w not all zero. If $w = 0$ then we reduce to the ternary case and there is nothing to prove, so suppose $w \neq 0$. Then we have

$$b = \frac{x^2 - ay^2}{z^2 - aw^2} = \frac{\text{Norm}_{k(\sqrt{a})/k}(x + \sqrt{a}y)}{\text{Norm}_{k(\sqrt{a})/k}(z + \sqrt{a}w)} = \text{Norm}_{k(\sqrt{a})/k}\left(\frac{x + \sqrt{a}y}{z + \sqrt{a}w}\right).$$

We now proceed with the proof of the special case:

Proposition 5: A special case of quaternary Hasse-Minkowski

Suppose $Q(x, y, z, w)$ is a quadratic form of square discriminant which is isotropic over all completions K_v of K . Then Q is isotropic over K .

Proof. Let $Q(x, y, z, w)$ be a quadratic form of square discriminant. As scaling by a constant does not affect isotropy, assume without loss of generality that $Q(x, y, z, w) = x^2 + ay^2 + bz^2 + cw^2$ and let $\text{disc}(Q) = abc = \alpha^2$. Recall that two quadratic forms Q_1 and Q_2 over K are equivalent iff there exists $C \in GL_n(K)$ such that $A_1 = C^T A_2 C$, where A_i is the defining matrix of Q_i . We can thus take $C = \text{diag}\left(1, \frac{\alpha}{a}, 1, 1\right)$ to see that Q is equivalent to the form $x^2 + \frac{\alpha^2}{a}y^2 + bz^2 + cw^2 = x^2 + bcy^2 + bz^2 + cw^2$ over each K_v . It follows by [lemma 1](#) that $x^2 + bz^2 + cw^2$ is isotropic over each K_v ; since Hasse-Minkowski holds for ternary forms, it follows that $x^2 + bz^2 + cw^2$ is isotropic over K . Then $x^2 + bcy^2 + bz^2 + cw^2$ is isotropic over K , as is $Q(x, y, z, w) = x^2 + ay^2 + bz^2 + cw^2$ by their equivalence. ■

To prove quaternary Hasse-Minkowski in its full generality, we will need the following fact:

Proposition 6

Let K be a field, $\alpha \in K^\times \setminus (K^\times)^2$, and let $L = K(\sqrt{\alpha})$. Suppose Q is a quadratic form which **is not** isotropic over K . Then the following are equivalent:

- (i) Q is isotropic over L .
- (ii) Q contains a binary subform f of discriminant $-\alpha$.

Proof. **(ii) $\Rightarrow$ (i):** Suppose f is a binary quadratic subform of Q with discriminant $-\alpha$. The transformation $\text{diag}\{1/\sqrt{\alpha}, 1\} \in GL_2(L)$ produces an equivalent form over L with discriminant -1 . Such a form is necessarily equivalent to $ax^2 + by^2$ where $ab = -1$; scaling by b gives $-x^2 + b^2y^2 = (x + by)(-x + by)$, which is isotropic over L . As f is a subform of Q , it follows that Q is isotropic over L .

(i) $\Rightarrow$ (ii): Let $Q = \sum_{i=1}^n a_i z_i^2$. Since Q is isotropic over L , there are $x_1, \dots, x_n, y_1, \dots, y_n \in K$ (not all zero) such that

$$\sum_{i=1}^n a_i (x_i + y_i \sqrt{\alpha})^2 = 0.$$

We split this equation into its K -rational and K -irrational parts to obtain

$$\begin{aligned}\sum_{i=1}^n a_i x_i^2 + \alpha \sum_{i=1}^n a_i y_i^2 &= 0 \\ \sum_{i=1}^n a_i x_i y_i &= 0.\end{aligned}$$

The second equation tells us that $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ are orthogonal in (K^n, Q) , while the first equation says that $Q(\mathbf{x}) = -\alpha Q(\mathbf{y})$. Since Q is not isotropic over K , it follows that $Q(\mathbf{x}), Q(\mathbf{y}) \neq 0$ and thus $\mathbf{x}, \mathbf{y} \neq \mathbf{0}$. Since $\alpha \notin (K^\times)^2$, $Q(\mathbf{x})$ and $Q(\mathbf{y})$ are in different classes in $K^\times / (K^\times)^2$. Thus $\mathbf{x}$ and $\mathbf{y}$ are a basis for a two-dimensional subspace of K^n on which Q restricts to the form $Q(\mathbf{x})x^2 + Q(\mathbf{y})y^2$, which has discriminant

$$Q(\mathbf{x})Q(\mathbf{y}) \equiv -\alpha \pmod{(K^\times)^2}.$$

Corollary 1

Let K be a field, Q a quaternary quadratic form over K with $\alpha := \text{disc}(Q)$. Let $L = K(\sqrt{\alpha})$; if Q is isotropic over L then Q is isotropic over K .

Proof. When $\alpha \in (K^\times)^2$, $L = K$ so the theorem is trivial. Suppose $\alpha \notin (K^\times)^2$ so $L \neq K$; suppose also for contradiction that Q is not isotropic over K . Since Q is isotropic over L , Q contains a binary subform f of discriminant $-\alpha$ by [proposition 6](#). As Q is quaternary, there is a binary subform g of Q with $\text{disc}(g) = -1$ ($Q = f \oplus g$). It follows from the discussion in the proof of [proposition 6](#) that g must be isotropic over K . This means Q must also be isotropic over K , contradicting the assumption that it isn't. ■

We finally have the ingredients necessary to prove quaternary Hasse-Minkowski in its entirety:

Proposition 7: Quaternary Hasse-Minkowski

The Hasse-Minkowski theorem holds for quaternary quadratic forms.

Proof. Suppose Q is isotropic over all completions K_v of K . Let $\alpha = \text{disc}(Q)$ and set $L = K(\sqrt{\alpha})$. By design, Q has square discriminant and is locally isotropic over all completions of L . [Proposition 5](#) implies Q is isotropic over L ; then Q is isotropic over K by [corollary 1](#). ■

2.4 $n \geq 5$

We need some lemmas relating to isotropy of quadratic forms over local fields, beginning with the following very special case of Chevalley's Theorem:

Lemma 2: (Chevalley)

Any (non-degenerate) quadratic form in at least three variables is isotropic over $\mathbb{F}_q$.

Proof. Suppose $a_1 x_1^2 + a_2 x_2^2$ is a non-degenerate two-dimensional quadratic form over $\mathbb{F}_q$. We have that

$$a_1 x_1^2 + a_2 x_2^2 = a_1 \left(x_1^2 + \frac{a_2}{a_1} x_2^2 \right).$$

If $-a_2/a_1 \in (\mathbb{F}_q^\times)^2$ then we have that the form is equivalent to $a_1(x_1 - x_2)(x_1 + x_2)$ and must be isotropic. If $-a_2/a_1 \notin (\mathbb{F}_q^\times)^2$, then we have

$$a_1x_1^2 + a_2x_2^2 = a_1 \text{Norm}_{\mathbb{F}_{q^2}/\mathbb{F}_q} \left(x_1 + \sqrt{\frac{-a_2}{a_1}} x_2 \right).$$

This form is not isotropic over $\mathbb{F}_q$; however since the norm map is surjective for finite fields, the form $a_1x_1^2 + a_2x_2^2$ must also be surjective onto $\mathbb{F}_q^\times$.

Now consider the form $a_1x_1^2 + \cdots + a_nx_n^2$ where $n > 2$. The subform $a_1x_1^2 + a_2x_2^2$ is either isotropic over $\mathbb{F}_q$ or surjective onto $\mathbb{F}_q^\times$. In either case, we can construct a nontrivial zero vector of the original form. ■

Theorem 8: Multivariate Hensel's Lemma

Let F be a nonarchimedean local field and $\mathcal{O}_F$ the associated valuation ring. If $f \in \mathcal{O}_F[x_1, \dots, x_n]$ and $\mathbf{a} \in \mathcal{O}_F^n$ satisfies

$$|f(\mathbf{a})| < \|(\nabla f)(\mathbf{a})\|^2$$

then there exists $\alpha \in \mathcal{O}_F^n$ such that $f(\alpha) = 0$ and $\|\alpha - \mathbf{a}\| < \|(\nabla f)(\mathbf{a})\|$.

Exercise 3

Prove **theorem 8**.

Lemma 3: A condition for local isotropy

Suppose Q is a non-degenerate diagonal quadratic form in at least three variables over a nonarchimedean local field F with odd residue characteristic. If each coefficient of Q is a unit in $\mathcal{O}_F$, then Q is isotropic over F .

Proof. Let π denote a uniformizer of $\mathcal{O}_F$, let κ denote the residue field $\mathcal{O}_F/\pi\mathcal{O}_F$, and let $q := |\kappa|$. By **lemma 2**, Q is isotropic over κ ; we can thus find $\mathbf{a} \in \mathcal{O}_F^n$ such that $f(\mathbf{a}) \equiv 0 \pmod{\pi}$. Further, we can guarantee that not all $a_i \equiv 0 \pmod{\pi}$, so it follows that $\|(\nabla f)(\mathbf{a})\|^2 \geq 1$. We thus have $|f(\mathbf{a})| \leq q^{-1} < 1 \leq \|(\nabla f)(\mathbf{a})\|^2$ which satisfies the conditions of **theorem 8**. Then there exists $\alpha \in \mathcal{O}_F^n$ such that $f(\alpha) = 0$ and $\|\alpha - \mathbf{a}\| < \|(\nabla f)(\mathbf{a})\|$. Note that since F has odd residue characteristic and each coefficient of Q is a unit in $\mathcal{O}_F$, we have $\|\mathbf{a}\| = \|(\nabla f)(\mathbf{a})\|$ so $\alpha \neq \mathbf{0}$. ■

Proposition 8: The Rest of Hasse-Minkowski

The Hasse-Minkowski Theorem holds for quadratic forms in at least five variables.

Proof. We will induct on the dimension of the form. The base cases have already been done, so suppose the theorem holds for quadratic forms of dimension $< n$ for some $n \geq 5$. Let $Q = Q_1 - Q_2$ where $Q_1 = a_1x_1^2 + a_2x_2^2$ and $Q_2 = -a_3x_3^2 - \cdots - a_nx_n^2$. Let S be the set of places of K containing all infinite places, all places over 2 (in the number field case), and all primes $\mathfrak{p}$ such that $v_{\mathfrak{p}}(a_i) > 0$ for some i . Note that S is necessarily finite. For all $\nu \in S$, Q_1 and Q_2 must represent a common value $\alpha_{\nu} \in K_{\nu}^\times$ as Q is isotropic over all K_{ν} . $(K_{\nu}^\times)^2$ is an open subgroup of $K_{\nu}^\times$, as is the coset containing α_{ν} . Since quadratic forms are continuous, the preimage of this coset under Q_1 (say A_{ν}) is also open in $K_{\nu} \times K_{\nu}$. The strong approximation theorem asserts that there exist $x_1, x_2 \in K$ such that $(x_1, x_2) \in A_{\nu}$ for all $\nu \in S$. Thus $\alpha := Q_1(x_1, x_2) \in K$ and $\alpha/\alpha_{\nu} \in (K_{\nu}^\times)^2$ for all $\nu \in S$. Then the form $\alpha y^2 - Q_2(x_3, \dots, x_n)$ is isotropic over K_{ν} for each $\nu \in S$; **lemma 3** implies that this form must also be isotropic over K_{ν} for all $\nu \notin S$. By the induction hypothesis, the form $\alpha y^2 - Q_2(x_3, \dots, x_n)$ is isotropic over K , so Q_2 represents α over K by **proposition 2**. This proves the theorem! ■

3 Number-Theoretic Applications

See also [4] (Appendix to Chapter IV) and chapter 1 of [3].

We now use the previous results on quadratic forms to prove some classic number-theoretic facts about representations of integers. In general, our results hold for quadratic forms over fields, so much of the obstruction to representability will lie in passing from representations in K to representations in $\mathcal{O}_K$. We begin with the following theorem:

Theorem 9: (Davenport-Cassels)

Let f be a quadratic form over $\mathbf{Q}$ defined by a symmetric matrix $A \in M_n(\mathbf{Z})$. Suppose that for any $\mathbf{x} \in \mathbf{Q}^n \setminus \mathbf{Z}^n$ there exists $\mathbf{y} \in \mathbf{Z}^n$ such that $0 < |f(\mathbf{x} - \mathbf{y})| < 1$. Then any $k \in \mathbf{Z}$ represented by f over $\mathbf{Q}$ is also represented by f over $\mathbf{Z}$.

Proof. Let $k \in \mathbf{Z}$ such that $f(\mathbf{x}) = k$ for some $\mathbf{x} \in \mathbf{Q}^n$. There exists $m \in \mathbf{Z}^+$ such that $m\mathbf{x} \in \mathbf{Z}^n$ so $m^2k = f(\mathbf{z})$ for some $\mathbf{z} \in \mathbf{Z}^n$. Let m and $\mathbf{z}$ be such that m is minimal; we will show that $m = 1$. By assumption, there exists $\mathbf{y} \in \mathbf{Z}^n$ such that

$$\frac{\mathbf{z}}{m} = \mathbf{y} + \mathbf{v} \quad \text{where } \mathbf{v} \in \mathbf{Q}^n \text{ such that } |f(\mathbf{v})| < 1.$$

If $f(\mathbf{v}) = 0$, then by assumption $\mathbf{v} = \mathbf{0}$ so $\frac{\mathbf{z}}{m} \in \mathbf{Z}^n$ meaning $m = 1$ by minimality. Suppose then that $f(\mathbf{v}) \neq 0$; set

$$a := f(\mathbf{y}) - k \quad b := 2(km - \langle \mathbf{z}, \mathbf{y} \rangle) \quad m' := am + b \quad \mathbf{w} := a\mathbf{z} + b\mathbf{y}.$$

By design, $a, b, m' \in \mathbf{Z}$ and

$$\begin{aligned} f(\mathbf{w}) &= \langle a\mathbf{z} + b\mathbf{y}, a\mathbf{z} + b\mathbf{y} \rangle \\ &= a^2 \langle \mathbf{z}, \mathbf{z} \rangle + 2ab \langle \mathbf{z}, \mathbf{y} \rangle + b^2 \langle \mathbf{y}, \mathbf{y} \rangle \\ &= a^2 m^2 k + ab(2km - b) + b^2(a + k) \\ &= k(a^2 m^2 + 2abm + b^2) \\ &= k(am + b)^2 \\ &= k(m')^2. \end{aligned}$$

Further,

$$\begin{aligned} mm' &= am^2 + bm \\ &= m^2 f(\mathbf{y}) - km^2 + 2km^2 - 2m \langle \mathbf{z}, \mathbf{y} \rangle \\ &= m^2 \langle \mathbf{y}, \mathbf{y} \rangle - 2m \langle \mathbf{z}, \mathbf{y} \rangle + \langle \mathbf{z}, \mathbf{z} \rangle \\ &= \langle m\mathbf{y} - \mathbf{z}, m\mathbf{y} - \mathbf{z} \rangle \\ &= f(-m\mathbf{v}) \\ &= m^2 f(\mathbf{v}) \end{aligned}$$

so $m' = mf(\mathbf{v})$. Since $0 < |f(\mathbf{v})| < 1$, we have $0 < |m'| < m$ which contradicts the minimality of m . Thus $f(\mathbf{v}) = 0$ so we are done. ■

Now we use this to prove the following classical representability statements:

Theorem 10: Two Squares (Fermat, Jacobi, Euler)

Let n be a positive integer. Then n can be written as the sum of two squares if and only if $v_p(n) \equiv 0 \pmod{2}$ for all primes $p \equiv 3 \pmod{4}$.

Proof. Without loss of generality, $n \not\equiv 0 \pmod{4}$. Indeed, n can be written as the sum of two squares if and only if $4n$ can be written as the sum of two squares. Similarly, we can assume $p^2 \nmid n$ for any prime $p \equiv 3 \pmod{4}$.

For $p \equiv 1 \pmod{4}$, there exists $\alpha \in \mathbf{Q}_p$ such that $\alpha^2 = -1$, so $x_1^2 + x_2^2$ is isotropic over $\mathbf{Q}_p$, meaning any $n \in \mathbf{Z}$ is represented by $x_1^2 + x_2^2$ over $\mathbf{Q}_p$. For odd p with $p \nmid n$, the form $x_1^2 + x_2^2 - nx_3^2$ is isotropic over $\mathbf{Q}_p$ by [lemma 3](#), so

n is represented by $x_1^2 + x_2^2$ over $\mathbf{Q}_p$ for such p . Since $n > 0$, the form $x_1^2 + x_2^2 - nx_3^2$ is isotropic over $\mathbf{R}$, so n is represented by $x_1^2 + x_2^2$ over $\mathbf{R}$ as well.

Now suppose $p \equiv 3 \pmod{4}$ is a prime dividing n (recall that by assumption, $p^2 \nmid n$). Then as -1 is not a quadratic residue mod p , we have that the only solution to $x_1^2 + x_2^2 \equiv 0 \pmod{p}$ is $x_1 \equiv x_2 \equiv 0 \pmod{p}$. But this implies that $x_1^2 + x_2^2 \equiv 0 \pmod{p^2}$, so we have that n cannot be a sum of squares in $\mathbf{Z}_p$ and thus in $\mathbf{Q}_p$. As n is not represented by $x_1^2 + x_2^2$ over $\mathbf{Q}_p$, n is also not represented by $x_1^2 + x_2^2$ over $\mathbf{Q}$.

Conversely, suppose p is not divisible by primes congruent to 3 mod 4. Then $x_1^2 + x_2^2 - nx_3^2$ is isotropic over $\mathbf{Q}_p$ for all odd primes p by **lemma 3**. It thus remains to show that n is represented by $x_1^2 + x_2^2$ over $\mathbf{Q}_2$. By assumption, n can be divisible only by 2 and primes congruent to 1 mod 4, and $4 \nmid n$. It follows that $n \equiv 1, 2$, or $5 \pmod{8}$. Now we can write $n = \alpha + \beta$ where $\alpha \in 1 + 8\mathbf{Z}_2$; then since $(\mathbf{Z}_2^\times)^2 = 1 + 8\mathbf{Z}_2$, we have $\alpha \in (\mathbf{Z}_2^\times)^2$. Further, we have $\beta = 0$, $\beta \in 1 + 8\mathbf{Z}_2$, or $\beta \in 4 + 8\mathbf{Z}_2$. In the former two cases, $\beta \in (\mathbf{Z}_2)^\times$, and in the latter case we write $\beta = 4\gamma$ where $\gamma \in 1 + 8\mathbf{Z}_2$, so $\beta \in (\mathbf{Z}_2)^\times$ in this case as well. Thus n is represented by $x_1^2 + x_2^2$ over $\mathbf{Q}_2$, so by **theorem 2**, n is represented by $x_1^2 + x_2^2$ over $\mathbf{Q}$. For $x_1, x_2 \in \mathbf{Q}$ (not both in $\mathbf{Z}$), we can find $y_1, y_2 \in \mathbf{Z}$ such that $|x_1 - y_1| \leq \frac{1}{2}$ and $|x_2 - y_2| \leq \frac{1}{2}$. As one of these must be nonzero, we have that $0 < (x_1 - y_1)^2 + (x_2 - y_2)^2 \leq \frac{1}{2} < 1$ so the form satisfies the conditions of **theorem 9**. Thus n is represented by $x_1^2 + x_2^2$ over $\mathbf{Z}$ as desired. ■

Theorem 11: Three Squares (Legendre, Gauss)

A positive integer n can be written as the sum of three squares if and only if $n = 4^k m$ where $(m, 4) = 1$ and $m \not\equiv 7 \pmod{8}$.

Proof. As with two squares, we assume that $4 \nmid n$, as n is the sum of three squares if and only if $4n$ is the sum of three squares. By **lemma 3**, the form $x_1^2 + x_2^2 + x_3^2$ is isotropic over $\mathbf{Q}_p$ for odd p , so n is represented by this form over $\mathbf{Q}_p$. Further, the form $x_1^2 + x_2^2 + x_3^2 - nx_4^2$ is isotropic over $\mathbf{R}$ since n is positive, so n is also a sum of three squares in $\mathbf{R}$. This reduces the theorem to proving when $\alpha \in \mathbf{Z}_2$ is a sum of three squares.

As $(\mathbf{Z}_2^\times)^2 = 1 + 8\mathbf{Z}_2$, we can repeat the arguments for **theorem 10** to see that n is a sum of three squares in $\mathbf{Z}_2$ if $n \equiv 1, 2, 3, 5$, or $6 \pmod{8}$. On the other hand, when $n \equiv 7 \pmod{8}$, we can see that n is not the sum of three squares mod 8, since the squares in $\mathbf{Z}_2/8\mathbf{Z}_2$ are 0, 1, 4. Thus, n is a sum of three squares in $\mathbf{Q}_2$ if and only if $n \not\equiv 7 \pmod{8}$; we note that this is also equivalent to $-n \notin (\mathbf{Z}_2)^\times$. Now by **theorem 2** and **theorem 9**, n is a sum of three squares in $\mathbf{Z}$ if and only if $n \not\equiv 7 \pmod{8}$ as desired. ■

Theorem 12: Four Squares (Lagrange)

Every positive integer n can be written as the sum of four squares in $\mathbf{Z}$.

Proof. We once again assume $4 \nmid n$; if the theorem is true for n then it is also true for $4n$. If $n \not\equiv 7 \pmod{8}$, then n is the sum of three squares and thus four squares in $\mathbf{Z}$. If $n \equiv 7 \pmod{8}$, then $n - 1 \equiv 6 \pmod{8}$ is a sum of three squares, so n is a sum of four squares. ■

Remark:

We note that it is possible to directly show, without reference to the result on three squares, that n is represented by $x_1^2 + x_2^2 + x_3^2 + x_4^2$ over $\mathbf{Q}_p$ for all $p \leq \infty$ (it is a fact that any quadratic form in at least five variables is isotropic over a complete DVR). However, this only guarantees a representation of n as a sum of four *rational* squares – **theorem 9** does not apply to this form.

Theorem 13: Polygonal Numbers (Fermat, Lagrange, Gauss, Cauchy)

Let n and $k \geq 2$ be nonnegative integers; we define the n^{th} k -gonal number

$$P_k(n) := \frac{1}{2} ((k-2)n^2 - (k-4)n).$$

Every positive integer m is the sum of k k -gonal numbers. The case $k = 3$ refers to triangular numbers and is known as Gauss' *Eureka Theorem*.

Proof. We note that the case $k = 2$ gives $P_2(n) = n$ and is trivial. The case $k = 4$ gives $P_4(n) = n^2$ and is done by Lagrange's theorem (**theorem 12**). For the case $k = 3$, let $m \in \mathbf{Z}^+$ and note that we can write $8m + 3 = x_1^2 + x_2^2 + x_3^2$ for some $x_1, x_2, x_3 \in \mathbf{Z}$ by Legendre's three squares theorem. Note also that since $0, 1, 4$ are the squares mod 8, we must have that $x_1 \equiv x_2 \equiv x_3 \equiv 1 \pmod{2}$, so write $x_i = 2y_i + 1$ for each i . Then

$$\sum_{i=1}^3 P_3(y_i) = \sum_{i=1}^3 \frac{y_i(y_i + 1)}{2} = \sum_{i=1}^3 \frac{1}{8} (4y_i^2 + 4y_i) = \frac{1}{8} \left[-3 + \sum_{i=1}^3 (2y_i + 1)^2 \right] = \frac{1}{8} \left[-3 + \sum_{i=1}^3 x_i^2 \right] = \frac{1}{8} [-3 + 8m + 3] = m.$$

For the proof of $k \geq 5$, we refer the reader to [3]. ■

4 A Cohomological Interpretation due to C. Lane

Let φ be an inner product on a finite dimensional k -vector space V , and let $O(K, \varphi)$ be the group of matrices with entries in K that preserve φ . Given two inner product spaces (V, φ) and (V', φ') , an isomorphism of inner product spaces is a map $T: V \rightarrow V'$ that commutes with the inner product, so $\varphi'(T\vec{v}, T\vec{w}) = \varphi(\vec{v}, \vec{w})$. We may also take a field extension K/k , and consider the K vector space $V_K = V \otimes K$. φ becomes an inner product φ_K on V_K .

For an extension of fields K/k and inner product space (V, φ) , we define the pointed set $E_\varphi(K/k)$ as the set of inner product spaces (V', φ') over k up to isomorphism that become isomorphic to (V, φ) when extended to K .

Theorem 14: Cohomology of the Orthogonal Group

Let K/k be an algebraic field extension, then there is a canonical isomorphism of pointed sets

$$E_\varphi(K/k) = H^1(K/k, O(\varphi, K)).$$

Where $H^1(K/k, O(\varphi, K))$ is a nonabelian cohomology group, which can be understood either as the pointed set of principal homogeneous spaces, or the pointed set of crossed homomorphisms modulo principal crossed homomorphisms.

Proof. We begin by defining the map. For a inner product space (V', φ') , let X be the set of isomorphisms $(V_K, \varphi_K) \rightarrow (V'_K, \varphi'_K)$. This is a principal homogeneous space, since it has a left action by $\text{Gal}(K/k)$ given by $\sigma(f) = \sigma \circ f \circ \sigma^{-1}$, and a transitive and faithful right action by $O(\varphi, K)$ given by $fT = f \circ T$.

We will show that this is a bijection by explicitly constructing its inverse $h: H^1(K/k, O(\varphi, K)) \rightarrow E_\varphi(K/k)$. Let X be a principal homogeneous space. For any $x \in X$, we can obtain a cocycle A_σ as the unique function $\text{Gal}(K/k) \rightarrow O(\varphi, K)$ such that $\sigma x = x A_\sigma$. Since $O(\varphi, K) \subseteq \text{GL}(V_K)$ and $H^1(K/k, \text{GL}(V_K)) = 1$, there is an $f \in \text{GL}(V_K)$ such that $A_\sigma = f^{-1} \sigma(f)$. Define φ' by $\varphi \circ (f, f)$. First, we show that φ' is a form over k . We compute

$$\sigma(\varphi') = \sigma(\varphi') \circ (\sigma(f), \sigma(f)) = \varphi' \circ (f \circ A_\sigma, f \circ A_\sigma) = \varphi \circ (f, f).$$

therefore φ' is defined over k and $(V, \varphi') \in E_\varphi(K/k)$.

We will now verify that these maps are inverses. Let $(V', \varphi') \in E_\varphi(K/k)$ be an inner product space over K . Let $f: (V_K, \varphi_K) \rightarrow (V'_K, \varphi'_K)$ be an isomorphism of inner product spaces. Selecting $x = f$ in the map from $H^1(K/k, O(\varphi, K))$ to $E_\varphi(K/k)$, we obtain the cocycle $A_\sigma = f^{-1} \sigma(f)$. It then follows by the definition of f that $\varphi'' = \varphi \circ (f, f)$ is equivalent

to φ . (There are minor technicalities here about the fact that f is technically not in $\mathrm{GL}(V)$ since it maps V_K to V'_K , but this issue is remedied by taking any isomorphism of vector spaces (not inner products) $\psi: V'_K \rightarrow V_K$ and considering φ'_K as acting on V_K in the way induced by ψ . Checking that they are inverses in the other direction is similar. ■

With this description of the cohomology of the orthogonal group, we motivate the study of when forms become equivalent. We need the following proposition, the proof of which is left as an exercise.

Exercise 4

Let φ and φ' be quadratic forms of the same dimension over an algebraically closed field $\bar{k}$. Show that φ and φ' are equivalent.

With this, we can now make a cohomological version of Hasse-Minkowski. Define the Tate-Shafarevich group of an group variety A defined over k by

$$\mathrm{III}(k, A) = \ker \left(H^1(k, A(\bar{k})) \rightarrow \prod_{\mathfrak{p}} H^1(k_{\mathfrak{p}}, A(\bar{k}_{\mathfrak{p}})) \right)$$

Note that when the group structure of A is nonabelian, this "kernel" is actually just the preimage of the distinguished element, and $1 = \{*\}$ is just the zero object in the category pointed sets.

Corollary 2

$$\mathrm{III}(k, O) = 1.$$

Proof. Let φ be a quadratic form over k , and O be the orthogonal group associated to it. Let φ' be any quadratic form over k . Then by **exercise 4** φ' and φ become equivalent over $\bar{k}$, so $\varphi' \in H^1(k, O(\bar{k}))$, but by **theorem 4** we have that φ' is equivalent to φ over k if and only if it is equivalent over all completions. In the language of cohomology groups, that means if $\varphi' = 1$ in $H^1(k_{\mathfrak{p}}, O(\bar{k}_{\mathfrak{p}}))$ for all primes $\mathfrak{p}$, then $\varphi' = 1$ in $H^1(k, O(k))$, so $\mathrm{III}(k, O) = 1$. ■

References

- [1] Emil Artin and John Tate. *Class Field Theory*. AMS Chelsea Publishing, 2009.
- [2] Pete L. Clark. Quadratic Forms over Global Fields. <http://alpha.math.uga.edu/~pete/qforms-global.pdf>.
- [3] Melvyn B. Nathanson. *Additive Number Theory: The Classical Bases*. Springer-Verlag, 1996.
- [4] Jean-Pierre Serre. *A Course in Arithmetic*. Springer-Verlag, 1973.